

DAVID SIERRA PORTA, PH.D.



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📍 **Actual Position:** Full Time Associate Professor (Physics and Data Science),
 Digital Transformation School and Basic Science Faculty,
 Universidad Tecnológica de Bolívar, UTB (utb.edu.co)
 📍 **Residency:** Cartagena de Indias, Colombia.

★ Professional Profile

- ▶ Director of three data science and applied statistics programs (undergraduate, master's, and specialization) at UTB.
- ▶ More than 15 years of university teaching experience in physics, mathematics, and data science.
- ▶ Senior Researcher (Minciencias) and peer reviewer; IEEE Senior Member; more than 50 indexed publications (Scopus/Scholar: h-index **11/18**, ~560/1040 citations).
- ▶ Previous academic ties to Bucaramanga: professorship and postdoctoral fellowship at UIS (2016–2022).

💡 Relevant Expertise

Throughout my career, I have developed solid and diverse experience in both **theoretical and experimental physics**, **advanced mathematics**, as well as in advanced data analysis and data science. My background in physics and mathematics has allowed me to tackle complex problems with a rigorous approach, applying nonlinear dynamic models and advanced resolution techniques to describe and solve physical and natural phenomena. This combination of theoretical and experimental expertise has provided me with a deep understanding of astroparticle fluxes, cosmic rays, and their interactions, as well as the characterization of these processes using specialized detectors. In addition to my experience in physics, I possess strong skills in **data science and advanced computational techniques**. I have applied **machine learning, deep learning, data mining, and artificial intelligence** methods to solve problems in multidisciplinary fields, developing predictive and descriptive models that address both scientific problems and challenges in industry and the financial sector. My ability to collect, organize, and analyze **large volumes of data** has enabled me not only to extract valuable insights but also to communicate these findings clearly and comprehensively, presenting the results in high-impact scientific journals and at international conferences. As a university professor with more than 15 years of experience, I have taught courses in areas as diverse as Quantum Mechanics, Statistical Physics, Statistics and Probability, Numerical Methods, and Data Science. My pedagogical approach focuses on guiding students through solving complex problems using mathematical and computational tools, providing them with a solid understanding that allows them to apply this knowledge across various disciplines. My active participation in the **management and execution of research projects** for both companies and universities has demonstrated my ability not only to lead teams but also to ensure that projects are successfully completed, delivering innovative and high-quality results. My competencies in preparing detailed reports and analyses have been key to effectively communicating progress and discoveries in every project I have been involved in.

🎓 Teaching — courses I can teach

- ▶ **Data Science and AI:** Machine Learning, Deep Learning, Data Mining, Artificial Intelligence, Scientific Visualization.
- ▶ **Statistics and Modeling:** Statistics and Probability, Computational Statistics, Time Series and Econometrics, Mathematical Modeling.
- ▶ **Mathematics:** Calculus (differential, integral, multivariable), Linear Algebra, Numerical Methods, Computational Tools.
- ▶ **Physics:** Physics I and II, Mechanics, Electromagnetism, Thermodynamics and Waves, Quantum Mechanics, Statistical Physics, Relativity.

🎓 Studies Completed - Academic Training

- 2016: **Ph.D. in Fundamental Physics.**
 Universidad de los Andes (ULA). Facultad de Ciencias. Mérida, Venezuela.
- Modified gravity studies and theories of gravity based on scalar, vector and tensor fields of higher order for the construction of alternative theories of gravity.
 - Quantum Field Theory - Gravitation and General Relativity.
 - Canonical analysis and duality analysis in field theories.
- 2004: **Magister Scientiae (M.Sc.) in Fundamental Physics.**
 Universidad de los Andes (ULA). Facultad de Ciencias. Mérida, Venezuela.
- Study of Chern-Simons super-gravity models.
 - Multi-dimensional super-gravity and quantum field theory.

- *Dimensional reduction in high-dimensional theories.*

2001: **Mathematics and Physics Degree.**

Universidad del Zulia (LUZ). Maracaibo, Venezuela.

- *Research assistant in mathematics and physics education projects in middle, high school and university education.*
- Assistant of the Laboratory for Physics Teaching of the Faculty of Humanities and Education.
- *Student Trainer and Teaching Assistant in the Department of Mathematics and Physics.*

1995: **Bachelor in Science.** Secondary and Bachelor of Science.

Colegio Marista San Pablo. Machiques, Venezuela.



Networks and Partnerships

- Member of International Astronomical Union: Paris, FR, 2024-06-21 to present.
- Member of American Physical Society (APS), No 62147697.
- Member of American Statistical Association (ASA), No 29579.
- Member of the INTER-AMERICAN STATISTICAL INSTITUTE (IASI).
- Member of the Colombian Statistical Society (SCE).
- Member of IEEE (IEEE Aerospace and Electronic Systems Society Membership).



Expertise - Work

2022-Today: **Full Time Associate Professor - Full Time.**

- Universidad Tecnológica de Bolívar. Digital Transformation School and Faculty of Basic Sciences. (<https://www.utb.edu.co/>).
- Researcher and Professor of the Master courses of Mechanics, Electromagnetism, Thermodynamics and Waves, Data Science, Machine Learning, Data mining, Time Series, Deep Learning.
- Researcher and Professor of Postgraduate Master Courses in Statistics and Data Science, Data Mining, Machine Learning, Time Series.
- Program Head of CIENCIA DE DATOS (Undergraduated), MAESTRÍA EN ESTADÍSTICA APLICADA Y CIENCIA DE DATOS (Master) y ESPECIALIZACIÓN EN ESTADÍSTICA APLICADA Y CIENCIA DE DATOS (Specialization).

2021-2022: **Senior Data Scientist.**

- Company: DBAccess (<https://dbaccess.com/>).
- Business Analytics. Data analytics projects for international insurance, contractors, investment, etc. companies. Decision management, modeling, forecasting, visualization for business.
- Information data collection, analyze and accurately interpret the information obtained, prepare reports and reports that allow the visualization of the data in an understandable way to a general public and other information products resulting from data analysis.

2021-2021: **Senior Data Analytics Expert.**

- Company: IMMAP-Colombia (<https://immap.org/colombia/>).
- Name of associated project: Migration of children and adolescents in Latin America, financed by UNICEF.
- To monitor the humanitarian situation of children, adolescents and their families in migration situations in Latin America and the Caribbean with a special focus on the conditions of mobility and access to basic services such as medical care, food and livelihoods, protection, education, shelter, water and sanitation, in order to inform programmatic decisions by UNICEF and its partners.
- Collect data information, analyze and accurately interpret the information obtained, prepare reports and briefs that allow the visualization of the data in a way that is understandable to a general audience and other information products resulting from the analysis of the data.

2021-2021: **Professor-Chair.**

- Science Faculty. Universidad Industrial de Santander. Physics Department. Bucaramanga, Santander - Colombia. (<https://www.uis.edu.co/>).
- Professor of Mathematical Modeling II (Graduate, Master's Degree in Applied Mathematics) and Computational Tools (Undergraduate).

2020-2022: **Fellow Research Postdoctoral Position.**

- Associate Project name: Dark Energy Spectroscopic Instrument (DESI).
- Physics Department. Universidad de los Andes, Colombia.
- Study and analysis of DESI project data <https://www.desi.lbl.gov/>. Active in working BGS (Bright Galaxy Survey) groups for data analysis.

2020-2022: **Master Professor.**

- Science Faculty. Universidad de los Andes. Physics Department. Bogotá, Colombia.
- Professor of the Master courses of Electromagnetism and Thermodynamics.

2019-2020: **Professor-Chair**

- Science Faculty. Universidad Industrial de Santander. Physics Department. Bucaramanga, Santander - Colombia.
 - Professor of Electromagnetism and Physics courses.
- 2018-2019: **Academic Staff - Research Associate.**
- Associate Project name: Research Associate on Muon Tomography.
 - Department of Physics and Astronomy. University of Sheffield. Sheffield, United Kingdom.
 - Study and analysis of astroparticle fluxes in volcanic structures for the application of the technique of Volcanic Muonography (Muon Tomography applied to Colombian volcanoes).
 - Theoretical and experimental studies in the field of astroparticles and characterization of cosmic ray detectors.
- 2017-2018: **Post-Doctoral Position.**
- Associate Project name: Muon Telescope (MuTe-UIS) for muonography of Colombian volcanoes.
 - Relativity and Gravitation Research Group (GIRG) and Halley Group of Astronomy and Aerospace Sciences. Industrial University of Santander. School of Physics. Bucaramanga, Colombia.
 - Study and analysis of astroparticle fluxes in volcanic structures for the application of the Volcanic Muonography technique (Muon tomography applied to Colombian volcanoes).
 - Theoretical and experimental studies in the field of astroparticles and characterization of cosmic ray detectors.
- 2016-2017: **Post-Doctoral Position.**
- Associate Project name: Simulation of scintillators and Cherenkov detectors for the muon telescope for volcanic muonography, MuTe-UIS.
 - Relativity and Gravitation Research Group (GIRG) and Halley Group of Astronomy and Aerospace Sciences. Industrial University of Santander. School of Physics. Bucaramanga, Colombia.
 - Study and analysis of astroparticle fluxes in volcanic structures for the application of the Volcanic Muonography technique (Muon tomography applied to Colombian volcanoes).
 - Theoretical and experimental studies in the field of astroparticles and characterization of cosmic ray detectors.
- 2016-2017: **Director of the Center for Scientific Modeling, CMC.**
- Experimental Science Faculty. Universidad del Zulia. Maracaibo, Venezuela.
- 2008-2018: **Ordinary Active Researcher of the Center for Scientific Modeling, CMC.**
- Experimental Science Faculty. Universidad del Zulia. Maracaibo, Venezuela.
 - Research in the area of Theoretical Physics and Relativity and Gravitation. Also researcher in the experiments for the study and characterization of the Relampago del Catatumbo, electroatmospheric discharges in the Eastern Coast of Maracaibo Lake, Venezuela.
- 2004-2015: **Ordinary Active Researcher of the Laboratory of Astronomy and Theoretical Physics, LAFT.**
- Experimental Science Faculty. Universidad del Zulia. Maracaibo, Venezuela.
- 2008-2018: **Full Time Professor-Researcher with Exclusive Dedication.**
- Experimental Science Faculty. Universidad del Zulia. Departamento de Física. Maracaibo, Venezuela.
 - A variety of courses have been taught: Classical Mechanics, Quantum Mechanics, Electromagnetism, General Physics 1, General Physics 2, Modern Physics, Statistics and Probability, Numerical Methods, Mathematical Methods I, Mathematical Methods II, Introduction to the Physics Laboratory, Experimental Techniques for Physicists.
- 2004-2008: **Regular Assistant Professor-Researcher with Part-time Dedication.**
- Mathematics and Physics Department. Humanities and Education Faculty. Universidad del Zulia.
 - A variety of courses have been taught including General Physics 1-5, Geometry, Linear Algebra, Calculus 1-3, Thermodynamics and Introduction to Physics Laboratory.
- 2004-2006: **Associate Professor Full Time.**
- Industrial Engineer School. Universidad Rafael Urdaneta (URU). Maracaibo, Venezuela.
 - A variety of courses have been taught including General Physics 1, Wave Mechanics, Thermodynamics, Statistics and Probability, and Introduction to the Physics Laboratory.
- 2005-2008: **Head of the Physics Teaching Laboratory.**
- Mathematics and Physics Department. Humanities and Education Faculty. Universidad del Zulia.
 - Management for the direction of the teaching laboratories of the School of Humanities and Education of the Universidad del Zulia.



Distinctions, recognitions and additional

- Recognized Peer Reviewer | Minciencias (Ministry of Science, Technology and Innovation)
Evaluation and issuance of technical and feasibility opinions for research, technological development, and innovation (R+D+I) projects in the areas of Science, Biotechnology, and Social Sciences.
- Senior Researcher (SR) | Minciencias (Ministry of Science, Technology and Innovation)
Highest category awarded nationally in recognition of high scientific output (Q1/Q2 articles, patents), continuous leadership of R+D+I projects, and training of advanced human capital (supervision of master's and doctoral theses).
- **IEEE Senior Member** | Institute of Electrical and Electronics Engineers

Highest professional grade for which IEEE members can apply, recognizing over 10 years of professional experience and at least 5 years of significant performance and technical leadership.

- **Director of Master Program: Applied Statistic and Data Science.**
Universidad Tecnológica de Bolívar. Basic Science Faculty. Cartagena de Indias, Colombia.
- **Co-creator of Undergraduate Program in Data Science.**
Universidad Tecnológica de Bolívar. Basic Science Faculty. Cartagena de Indias, Colombia.
- Director of the Semillero de Astronomía y Ciencia de Datos of the Universidad Tecnológica de Bolívar. Faculty of Basic Sciences. Cartagena de Indias, Colombia.
- Director/Leader of the Research Group in Gravitation and Applied Mathematics of the Universidad Tecnológica de Bolívar. Faculty of Basic Sciences. Cartagena de Indias, Colombia.
- Lecture Courses in International Workshop in Applied Statistic and data Science en Cartagena de Indias, Colombia, based on Data Mining and una Introduction to Machine learning.
- Lecture Courses at El Encuentro Matemático del Caribe in Cartagena de Indias, Colombia, about A gentle Introduction to Gradient Descent in Data Science.
- Organizer of the International Workshop in Applied Statistic and Data Science en Universidad Tecnológica de Bolívar, Basic Science Faculty, Cartagena de Indias, Colombia. The event is realized each year in June.

Research Projects

1. Observación solar, clima espacial y ciencia de datos. Medición de la rotación solar a partir de imágenes de cámaras digitales imágenes. David Sierra Porta (Principal Investigator), Yaleidys Paola Hernández Diaz (Co-Principal Investigator). UTB Research Division Internal Call for Proposals. **(\$35,900,000)**. Grant **INV03CI2205**. Start: may 17, 2022, Finish: january 17, 2023. **Financed**.
2. Cartografía de los estudios migratorios: Análisis de Fuentes Secundarias sobre la Migración en América Latina Y El Caribe a partir de metodologías de Ciencias de Datos. Andy Domínguez Monterrosa (Principal Investigator), David Sierra Porta (Co-Principal Investigator). UTB Research Division Internal Call for Proposals. **(\$35,900,000)**. Grant **INV03CI2205**. Start: may 17, 2022, Finish: january 17, 2023. **Financed**.
3. Observación solar, clima espacial y ciencia de datos. Estudio estadístico de los parámetros de actividad solar del ciclo solar 24. Jorge Villalba Acevedo (Principal Investigator), David Sierra Porta (Co-Principal Investigator). UTB Research Division Internal Call for Proposals. **(\$35,900,000)**. Grant **INV03CI2205**. Start: may 17, 2022, Finish: january 17, 2023. **Financed**.
4. Simetría de dualidad para teorías de espín 2 masivas. Principal Investigator. Financiado por la División de Investigación de la Facultad Experimental de Ciencias de la Universidad del Zulia, bajo el código FDI-12-2015.
5. Soluciones de la ecuación de Thomas-Fermi con aproximación de métodos de cálculo variacional. Investigador Principal. Financiado por la División de Investigación de la Facultad Experimental de Ciencias de la Universidad del Zulia, bajo el código RDI-351-2015.
6. Algunos aspectos sobre la dualidad gravitacional. Co-Principal Investigator. Funded by the Research Division of the Experimental Faculty of Sciences of the Universidad del Zulia, under the code RDI-352-2015.
7. Radiación de Cargas Aceleradas. Estudio de la Ecuación de Lorentz-Dirac. Co-Investigador. Funded by the Research Division of the Experimental Faculty of Sciences of the Universidad del Zulia, under the code FDI-25-20010.
8. Cosmología con Campos Escalares y Ecuación de Estado No-Local. Principal Investigator. Funded by the Research Division of the Experimental Faculty of Sciences of the Universidad del Zulia, under the code FDI-30-2009.
9. Estudio Canónico-Dinámico de la Gravedad cuatro (4) dimensional con simetría transversa de Fierz-Pauli. Principal Investigator and Responsible. Registered by the Council of Scientific and Humanistic Development, CONDES, under the code CC-0689-2008.
10. Cosmología con Ecuación de Estado No-Local. Principal Investigator and Responsible. Registered by the Council of Scientific and Humanistic Development, CONDES, under the code CC-0844-2008.
11. Formulación Variacional de Ecuaciones Diferenciales Parciales Estocásticas. Co-Principal Investigator and Responsible. Registered by the Council of Scientific and Humanistic Development, CONDES, under the code DI-FEC-702-2004.

Publications in peer-reviewed journals

Research Areas

1. **Applied Data Science and Artificial Intelligence:** Modeling of social, economic, and financial problems; analytics and predictive models for insurance, business, and decision-making.
2. **Space Weather and Solar Physics:** Solar dynamics and its influence on cosmic rays, using complex network methods, discrete curvature, and multifractal time series analysis.
3. **Astroparticle Physics and Muography:** Muon tomography applied to Colombian volcanic structures.
4. **Mathematical Physics:** Methods for differential equations with a high degree of nonlinearity; approximate and exact solutions.

Peer-reviewed journals

See also:

 GOOGLE SCHOLAR: <https://scholar.google.com/citations?user=-OInFfYAAAAJ&hl=en>

 ORCID: <https://www.orcid.org/0000-0003-3461-1347>

 SCOPUS author ID: 57191333650: <https://www.scopus.com/authid/detail.uri?authorId=57191333650>

1. Porta, D. S. (2026). Cayley–Menger curvature: a geometric approach to discrete curvature on graphs via local simplex embedding. *Journal of Complex Networks*, 14(4), cnago28. <https://doi.org/10.1093/comnet/cnag028>.
2. Porta, D. S. (2026). A Data-Driven Beacon of Hope: Review of Hannah Ritchie. 2024. *Not the End of the World: How We Can Be the First Generation to Build a Sustainable Planet*. Chatto & Windus, Penguin Random House UK. *Ecology, Economy and Society—the INSEE Journal*, 9(2), 1693-1. <https://doi.org/10.37773/ees.v9i2.1693>.
3. Sierra-Porta, D., & Varotsos, C. (2026). Heliospheric regimes governing Forbush decreases revealed by multiscale analysis of cosmic rays. *Acta Astronautica*. Volume 249, Part A. <https://doi.org/10.1016/j.actaastro.2026.07.024>.
4. Navarro, J. D. P., & Porta, D. S. (2026). Morphological Fingerprints of Forbush Decreases and Their Relation to Geomagnetic Storm Severity. *The Open Journal of Astrophysics*, 9. <https://doi.org/10.33232/001c.164441>.
5. Sierra-Porta, D. (2026). Magnitude-dependent quantum advantage in Forbush decrease detection: A quantum kernel SVM benchmark. *Astronomy and Computing*, 57(1), 101160. <https://doi.org/10.1016/j.ascom.2026.101160>.
6. Sierra Porta, D., Díaz Ramírez, O., & Tobón Ospino, M. (2026). Tourism resilience from networks: diversity and hierarchy. *Journal of Complex Networks*, 14(3), cnago21. <https://doi.org/10.1093/comnet/cnag021>.
7. Sierra-Porta, D., Arrieta-Guardo, I. S., Gonzalez-Ortiz, J. F., & Negrin-Perez, E. A. (2026). A Nonlinear, Robust Station–Season Index for Quantifying Fine–Coarse Mixture Regimes in Particulate Matter. *CLEAN–Soil, Air, Water*, 54(5), e70191. <https://doi.org/10.1002/clen.70191>.
8. Quintero-Pareja, Fabian C., Diederik A. Montano-Burbano, Santiago Quintero-Pareja, and David Sierra Porta. 2026. “Vision-Based CNN Prediction of Sunspot Numbers from SDO/HMI Images.” *The Open Journal of Astrophysics* 9 (March). <https://doi.org/10.33232/001c.159191>.
9. Petro-Ramos, Jesús D., David J. Ruiz-Morales, and David Sierra Porta. 2026. “Graph-Based Light-Curve Features for Robust Transient Classification.” *The Open Journal of Astrophysics* 9 (March). <https://doi.org/10.33232/001c.159506>.
10. Perez-Navarro, Juan D., and David Sierra Porta. 2026. “Derivative-Aligned Anticipation of Forbush Decreases from Entropy and Fractal Markers.” *The Open Journal of Astrophysics* 9 (February). <https://doi.org/10.33232/001c.157585>.
11. Sierra-Porta, D., Verdugo, M. C., & Acevedo, D. D. H. (2026). Complexity and scaling descriptors as diagnostic predictors of heliophysical indices across solar-cycle timescales. *Advances in Space Research*. Volume 77, Issue 6, 15 March 2026, Pages 6645-6658. <https://doi.org/10.1016/j.asr.2026.02.010>.
12. Ospino, M. T., Peña, H. M., & Porta, D. S. (2026). A Global and Hemispheric Analysis of the Relationship between Environmental Sustainability and Economic Inequality. *Ecology, Economy and Society—the INSEE Journal*, 9(1), 87-119. <https://doi.org/10.37773/ees.v9i1.1691>.
13. Sierra-Porta, D. (2025). Scale-dependent coupling between galactic cosmic rays and trace gases revealed by multifractal analysis. *Journal of Atmospheric and Solar-Terrestrial Physics*, 277, 106661. <https://doi.org/10.1016/j.jastp.2025.106661>.
14. Sierra-Porta, D., Verdugo, M. C., & Acevedo, D. D. H. (2025). Global complexity signatures of solar cycles: A unified Entropy–Fractal survey of OMNI solar wind data (1964–2025). *Advances in Space Research*. Volume 76, Issue 11, 1 December 2025, Pages 7190-7204. <https://doi.org/10.1016/j.asr.2025.09.072>.
15. Herrera-Acevedo, D. D., & Sierra-Porta, D. (2025). Urban mobility insights: A dataset for exploring network topology and city dynamics. *Data in Brief*, 112076. <https://doi.org/10.1016/j.dib.2025.112076>.
16. Figueroa, M. G., Acevedo, D. D. H., & Porta, D. S. (2025). Geomagnetic disturbances and grid vulnerability: Correlating storm intensity with power system failures. *PLoS One*, 20(7), e0327716. <https://doi.org/10.1371/journal.pone.0327716>.
17. Yopez, D. B., Porta, D. S., Aguas, L. M., & Jattin, F. M. (2025). Optimizing treatment to control LDL cholesterol using machine learning. *Computers in Biology and Medicine*, 195, 110599. <https://doi.org/10.1016/j.compbiomed.2025.110599>.
18. Sierra-Porta, D. (2025). Characterizing long-term cosmic ray time series with geometric network curvature metrics. *Journal of Atmospheric and Solar-Terrestrial Physics*, 268, 106418. <https://doi.org/10.1016/j.jastp.2025.106418>.
19. Herrera-Acevedo, D. D., & Sierra-Porta, D. (2025). Network structure and urban mobility sustainability: A topological analysis of cities from the urban mobility readiness index. *Sustainable Cities and Society*, 119, 106076. <https://doi.org/10.1016/j.scs.2024.106076>.
20. Sierra-Porta, D. (2025). Revised cross-correlation and time-lag between cosmic ray intensity and solar activity using Chatterjee’s correlation coefficient. *Advances in Space Research*, 75(1), 1330-1342. <https://doi.org/10.1016/j.asr.2024.10.065>.
21. Sierra-Porta, D. (2025). Multifractal detrended cross-correlation coefficient for cosmic ray and sunspot time series. *Journal of Atmospheric and Solar-Terrestrial Physics*, 266, 106407. <https://doi.org/10.1016/j.jastp.2024.106407>.
22. Tarazona Alvarado, M., Salamanca-Coy, J. L., Forero-Gutiérrez, K., Núñez, L. A., Pisco-Guabave, J., Escobar-Díaz, F., & Sierra-Porta, D. (2024). Assessing and monitoring air quality in cities and urban areas with a portable, modular and low-cost sensor station: calibration challenges. *International Journal of Remote Sensing*, 45(17), 5713-5736. <https://doi.org/10.1080/01431161.2024.2373338>.
23. D. Sierra-Porta, J.D. Petro-Ramos, D.J. Ruiz-Morales, D.D. Herrera-Acevedo, A.F. García-Teheran, M. Tarazona Alvarado (2024). Machine learning models for predicting geomagnetic storms across five solar cycles using Dst index and heliospheric variables. *Advances in Space Research*, Volume 74, Issue 8, Pages 3483-3495. <https://doi.org/10.1016/j.asr.2024.08.031>.
24. Sierra-Porta, D. (2024). A multifractal approach to understanding Forbush Decrease events: Correlations with geomagnetic storms and space weather phenomena. *Chaos, Solitons & Fractals*, 185, 115089. <https://doi.org/10.1016/j.chaos.2024.115089>.
25. Sierra-Porta, D., Tarazona-Alvarado, M., & Acevedo, D. H. (2024). Predicting sunspot number from topological features in spectral images I: Machine learning approach. *Astronomy and Computing*, 48, 100857. <https://doi.org/10.1016/j.ascom.2024.100857>.

26. Sierra-Porta, D. (2024). Assessing the impact of missing data on water quality index estimation: a machine learning approach. *Discover Water*, 4(1), 11. <https://doi.org/10.1007/s43832-024-00068-y>.
27. Sierra-Porta, D. (2024). Relationship between magnetic rigidity cutoff and chaotic behavior in cosmic ray time series using visibility graph and network analysis techniques. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 34(2). <https://doi.org/10.1063/5.0167156>.
28. Tarazona-Alvarado, M., & Sierra-Porta, D. (2023). Dataset for Sun dynamics from topological features. *Data in Brief*, 51, 109728. <https://doi.org/10.1016/j.dib.2023.109728>.
29. Porta, D. S., Acevedo, D. H., Tarazona-Alvarado, M., & Díaz, Y. H. (2023). SunspotCalc: Una aplicación basada en Web y Python para calcular la rotación diferencial del sol y su fotosfera. *Revista Mexicana de Física E*, 20(2 Jul-Dec), 020208-1. <https://doi.org/10.31349/RevMexFis.20.020208>.
30. Hahn, C., Wilson, M. J., Ruiz-Macias, O., Cole, S., Weinberg, D. H., Moustakas, J., ... & Zou, H. (2023). The DESI Bright Galaxy Survey: Final Target Selection, Design, and Validation. *The Astronomical Journal*, 165(6), 253. <https://doi.org/10.3847/1538-3881/acff8>.
31. Sierra-Porta, D., Solano-Correa, Y.T., Tarazona-Alvarado, M. and de Villaviscencio, L.A.N., Linking PM10 and PM2.5 Pollution Concentration Through Tree Coverage in Urban Areas. *CLEAN–Soil, Air, Water*, 51, 5 2200222 (2023). <https://doi.org/10.1002/clen.202200222>.
32. Sierra-Porta, D., Tarazona-Alvarado, M., & Villalba-Acevedo, J. (2023). Quantitatively relating cosmic rays intensities from solar activity parameters based on structural equation modeling. *Advances in Space Research*, 72(2), 638–648. <https://doi.org/10.1016/j.asr.2023.02.044>.
33. Lan, T. W., Tojeiro, R., Armengaud, E., Prochaska, J. X., Davis, T. M., Alexander, D. M., ... & Zhou, Z. (2023). The DESI Survey Validation: Results from Visual Inspection of Bright Galaxies, Luminous Red Galaxies, and Emission-line Galaxies. *The Astrophysical Journal*, 943(1), 68. <https://doi.org/10.3847/1538-4357/aca5fa>.
34. Sierra-Porta, D. (2022). On the fractal properties of cosmic rays and Sun dynamics cross-correlations. *Astrophysics and Space Science*, 367(12), 1–14. <https://doi.org/10.1007/s10509-022-04151-5>.
35. de León-Barrios R., Peña-Rodríguez J., Sanabria-Gómez J.D., Vásquez-Ramírez A., Calderón-Ardila R., Sarmiento-Cano C., Vesga-Ramírez A., Sierra-Porta D., Suárez-Durán M., Asorey H., Núñez L.A. (2022). Muography for the Colombian. *Proceedings of Science*, 395, art. no. 280.
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Books

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2. Juego y Aprendo a Calcular. Carlos Eduardo Guédez Torrez & David Sierra-Porta. Editorial Fe y Alegría (2006-05-04). ISBN 13978-980-6418-79-0, ISBN 10980-6418-79-4.



Thesis Tutoring-Mentoring

1. Director of Master's Thesis: Análisis de reseñas de TripAdvisor sobre el Castillo de San Felipe y la Ciudad Amurallada de Cartagena mediante minería de datos y aprendizaje automático. Maestría en Estadística Aplicada y Ciencia de Datos de la Universidad Tecnológica de Bolívar (2025). Thesis writer: Gabriel Enrique Calvo Guzman.
2. Director of Master's Thesis: Sistema de Inteligencia Comercial basado en Machine Learning para la identificación y recomendación de mercados internacionales con demanda insatisfecha para la oferta exportable del departamento de Bolívar. Maestría en Estadística Aplicada y Ciencia de Datos de la Universidad Tecnológica de Bolívar (2025). Thesis writer: Gustavo Enrique López Galvis.
3. Director of Master's Thesis: Relación Entre Los Indicadores De Salud Mental Y Los Objetivos De Desarrollo Sostenible: Un Análisis De Los Perfiles De Los Países Mediante Técnicas De Machine Learning. Maestría en Estadística Aplicada y Ciencia de Datos de la Universidad Tecnológica de Bolívar (2025). Thesis writer: Mildred Lucia Caro Álvarez.
4. Director of Master's Thesis: Resiliencia turística en Colombia ante choques exógenos: un enfoque desde la ciencia de redes, la entropía y la fractalidad con apoyo de aprendizaje profundo. Maestría en Estadística Aplicada y Ciencia de Datos de la Universidad Tecnológica de Bolívar (2025). Thesis writer: Oscar Javier Díaz Ramírez.
5. Director of Master's Thesis: Análisis de la relación entre condiciones socioeconómicas y diagnóstico del cáncer de mama en mujeres de Sincelejo sucre. Maestría en Estadística Aplicada y Ciencia de Datos de la Universidad Tecnológica de Bolívar (2025). Thesis writer: Sindy Milena Velasquez.
6. Director of Master's Thesis: Metodología para la construcción de líneas base energéticas de consumo absoluto horario utilizando algoritmos de agrupación estandar. Maestría en Estadística Aplicada y Ciencia de Datos de la Universidad Tecnológica de Bolívar (2025). Thesis writer: Luis Fernando Flórez García.

7. Director of Master's Thesis: Pronóstico de la tasa de cambio representativa del mercado colombiano (TRM) con redes neuronales recurrentes LSTM. Maestría en Estadística Aplicada y Ciencia de Datos de la Universidad Tecnológica de Bolívar (2024). Thesis writer: Yuleidis Mesa González.
8. Director of Master's Thesis: Nivel de procrastinación en estudiantes de la Universidad Tecnológica de Bolívar. Maestría en Estadística Aplicada y Ciencia de Datos de la Universidad Tecnológica de Bolívar (2024). Thesis writer: Rafael David Cueto Rodríguez.
9. Director of Master's Thesis: Análisis de Impacto de la Estrategia ALEKS en el Rendimiento Académico de Estudiantes de la Universidad Tecnológica de Bolívar: Una Evaluación en el Contexto de Cálculo Diferencial. Maestría en Estadística Aplicada y Ciencia de Datos de la Universidad Tecnológica de Bolívar (2024). Thesis writer: Humberto Manuel Marbello Pena.
10. Director of Master's Thesis: Revelando complejas interacciones entre el Índice de Desarrollo Humano y los Objetivos de Desarrollo Sostenible: Un análisis basado en el aprendizaje automático. Maestría en Estadística Aplicada y Ciencia de Datos de la Universidad Tecnológica de Bolívar (2024). Thesis writer: Hugo Alberto Forero Guerra.
11. Director of Master's Thesis: Optimización del tratamiento para controlar el colesterol LDL usando modelos de inteligencia artificial. Maestría en Estadística Aplicada y Ciencia de Datos de la Universidad Tecnológica de Bolívar (2024). Thesis writer: Deiby Jhon Boneu Yopez.
12. Co-Director of Master's Thesis: Inversion geofísica a partir de datos de muografía volcanica para proyecto MuTe. Maestría en Geofísica. Universidad Industrial de Santander. 2016. Thesis writer: Maria Alejandra Vesga Ramirez.
13. Undergraduate Thesis Tutor: Soluciones semi-exactas a la Ecuación de Thomas-Fermi. Thesis writer: Br. María Chirinos. Licenciado en Física. Facultad Experimental de Ciencias. La Universidad del Zulia. Mayo de 2016.
14. Master's Thesis Tutor: Soluciones esféricamente simétricas de las ecuaciones de Einstein + términos no-polinomiales y su efecto en la geometría del espacio-tiempo. Thesis writer: Guerrero Ruíz, Jefferson José. Magister Scientiarum en Física. Facultad Experimental de Ciencias. La Universidad del Zulia. Enero de 2011.
15. Master's Thesis Tutor: Cosmología con campos escalares y ecuación de estado Nlocal. Thesis writer: José Gerardo Caldera. Magister Scientiarum en Física. Facultad Experimental de Ciencias. La Universidad del Zulia. Marzo de 2010.
16. Master's Thesis Tutor: Estudio teórico de un péndulo esférico bajo la acción de un campo magnético tipo monopolo. Thesis writer: Montiel Cubillan, Germain Andrés. División de Estudios de Postgrado de la Facultad de Ingeniería. Universidad del Zulia. Postgrado en Física Aplicada. Facultad de Ingeniería. Diciembre 2007.



Interests

- **Data Science, Computational Tools of Data Science, Robust algorithms for Artificial Intelligence, Machine Learning, Deep Learning, Data Mining, Scientific Visualization.**
- I am particularly interested in atmospheric dynamics, air quality, and water quality, and how data science can be applied to evaluate, describe, and study these processes.
- I am deeply interested in solar dynamics, the influence of the solar wind, and heliospheric conditions on Earth, including cosmic rays and low- and high-energy astroparticles. I also explore the various applications of cosmic rays in scientific research.
- Mathematical Physics, Mathematical Methods, Experimental Physics, Modeling of Natural Phenomena, Data Science, Data Analysis, General Relativity. Gravitation and Cosmology. Analytical Mechanics.
- Teaching at middle school, high school, diversified, professional, secondary and high school levels. Teaching and research in university higher education.
- Outdoor sports, running, cycling, etc.



Skills

Computer Science PYTHON, R, MAPLE, MATHEMATICA SOFTWARE, SPSS, PSPP, ORIGIN, LATEX, OVERLEAF, OPENOFFICE, EXCEL-MICROSOFT, WORD-MICROSOFT, LINUX, MICROSOFT WINDOWS, GIMP, KADABRA.

Communication More than 20 publications in indexed and refereed journals (publications attached). Oral presentation in congresses and events through conferences.

Teaching at different levels of secondary and university education. Group management and excellent command of skills for communication and administration of learning processes.

Experimental and Theoretical Physics and Mathematics. Experience in tensor calculus and techniques in General Relativity and Cosmology. Study of systems and theories in Quantum Field Theory in several dimensions. Study of systems in the context of theoretical mechanics and theoretical physics in general.



Languages

Spanish: Native proficiency, Native Language

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Professional References

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